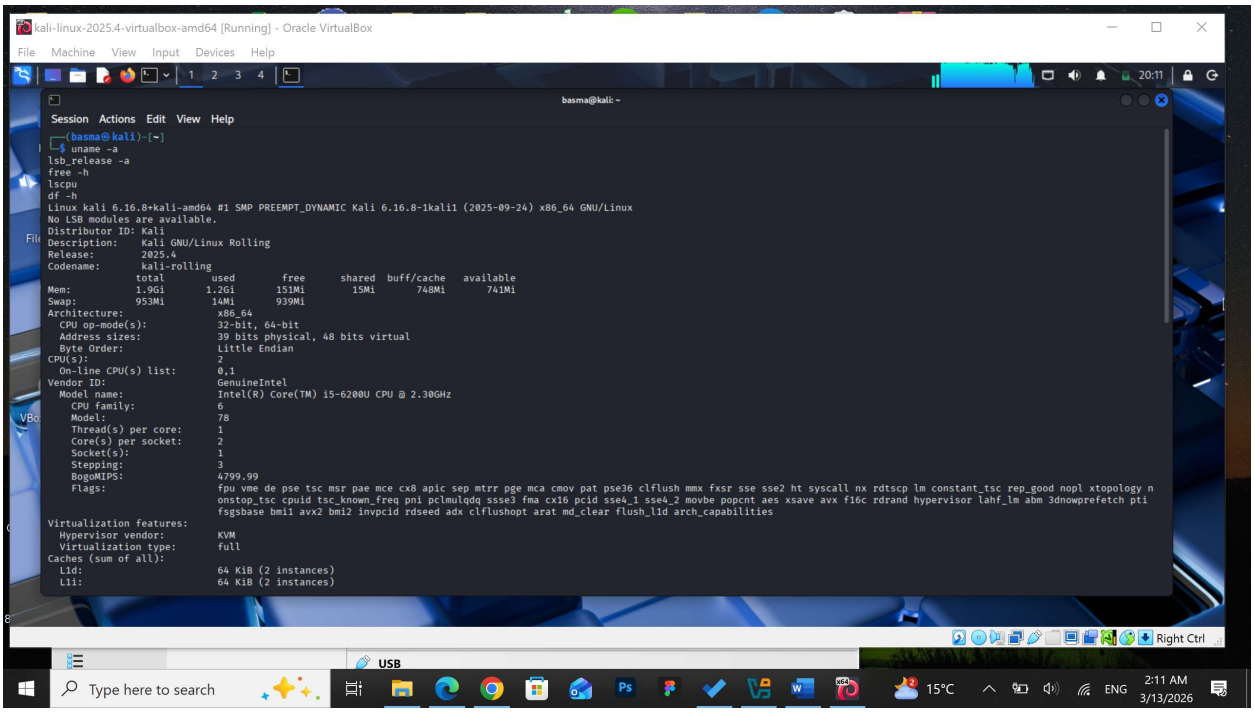
Name: Basma Mohamed Mostafa

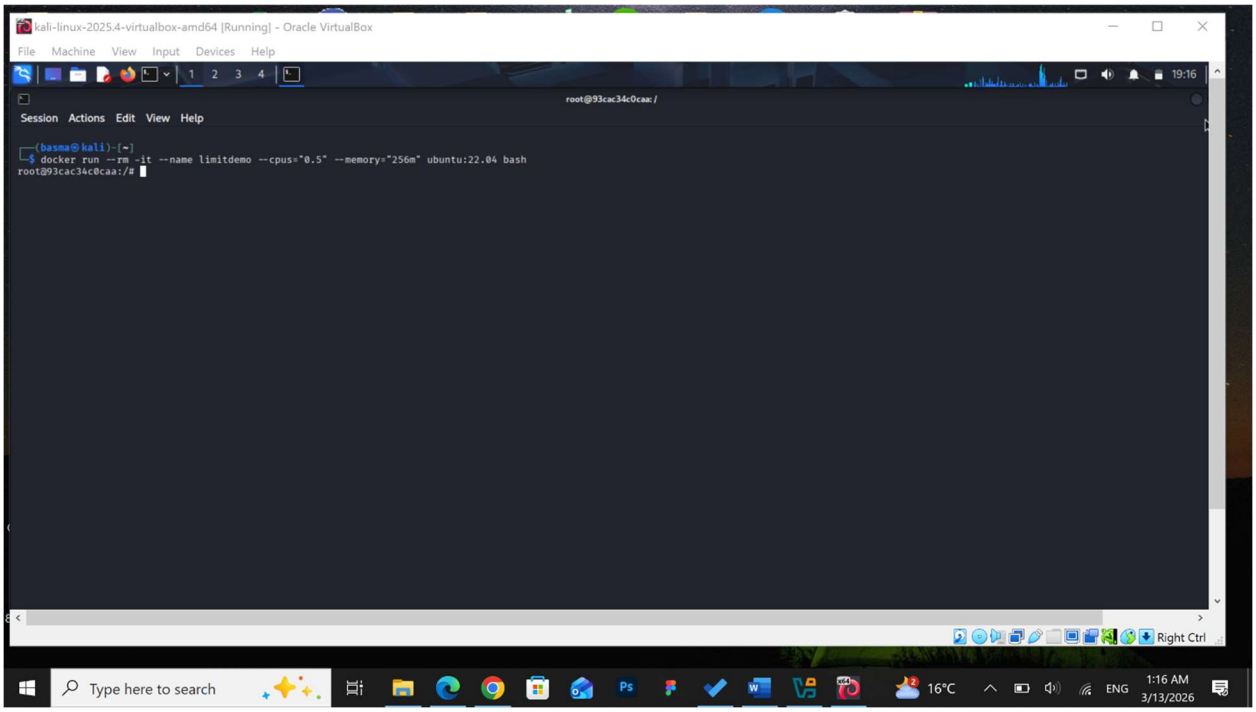
Group: SE3

Lab 3

Device Properties:

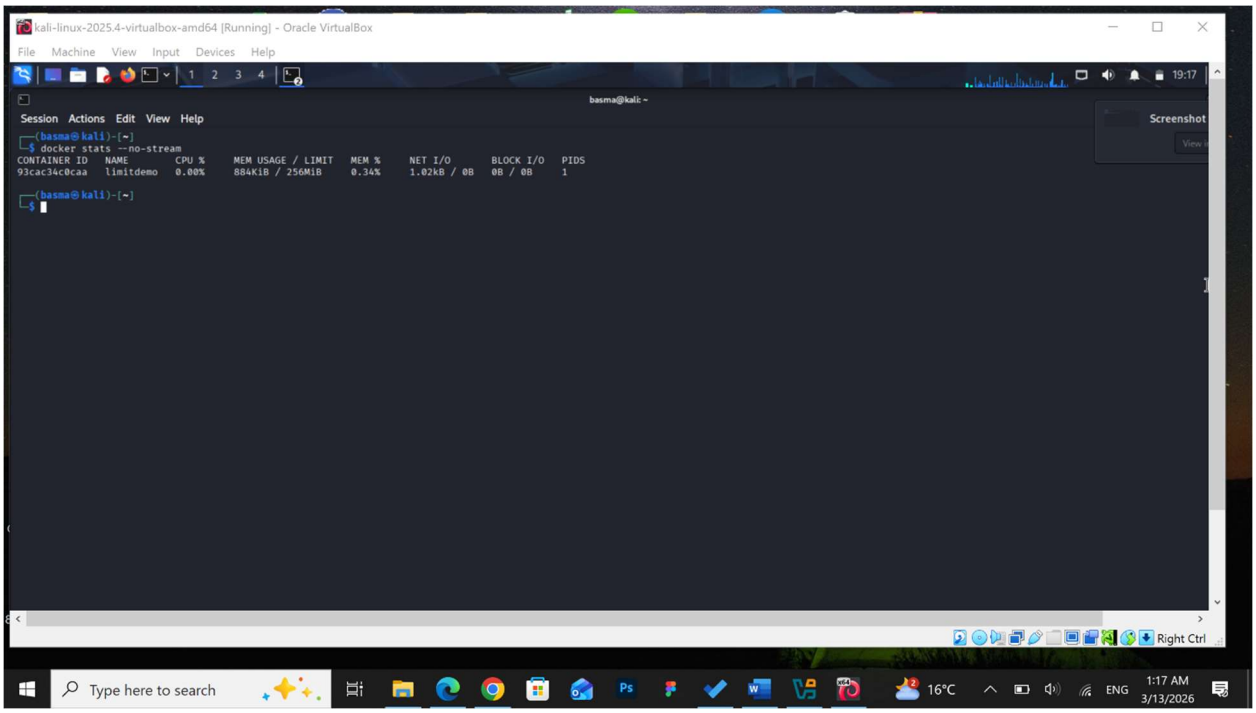


1. Start a resource-limited container

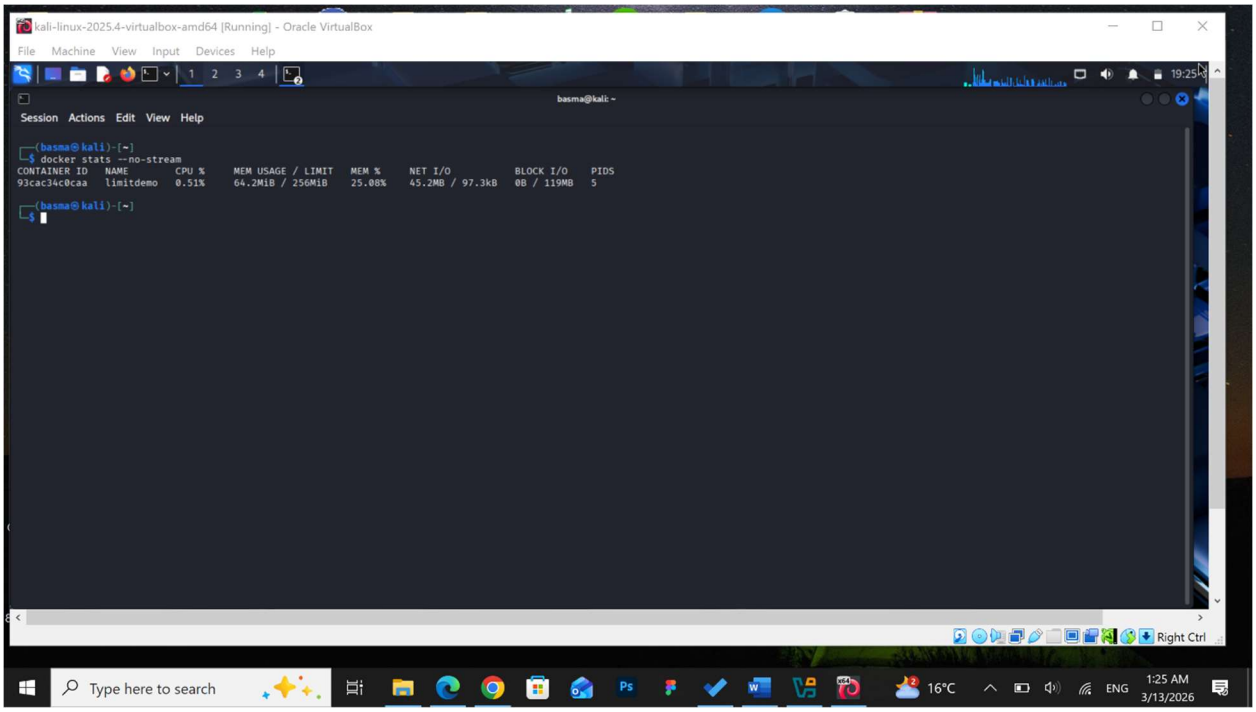


2. قبل وبعد تنفيذ الـ Stress Container مع عرض حالة ال

Before:



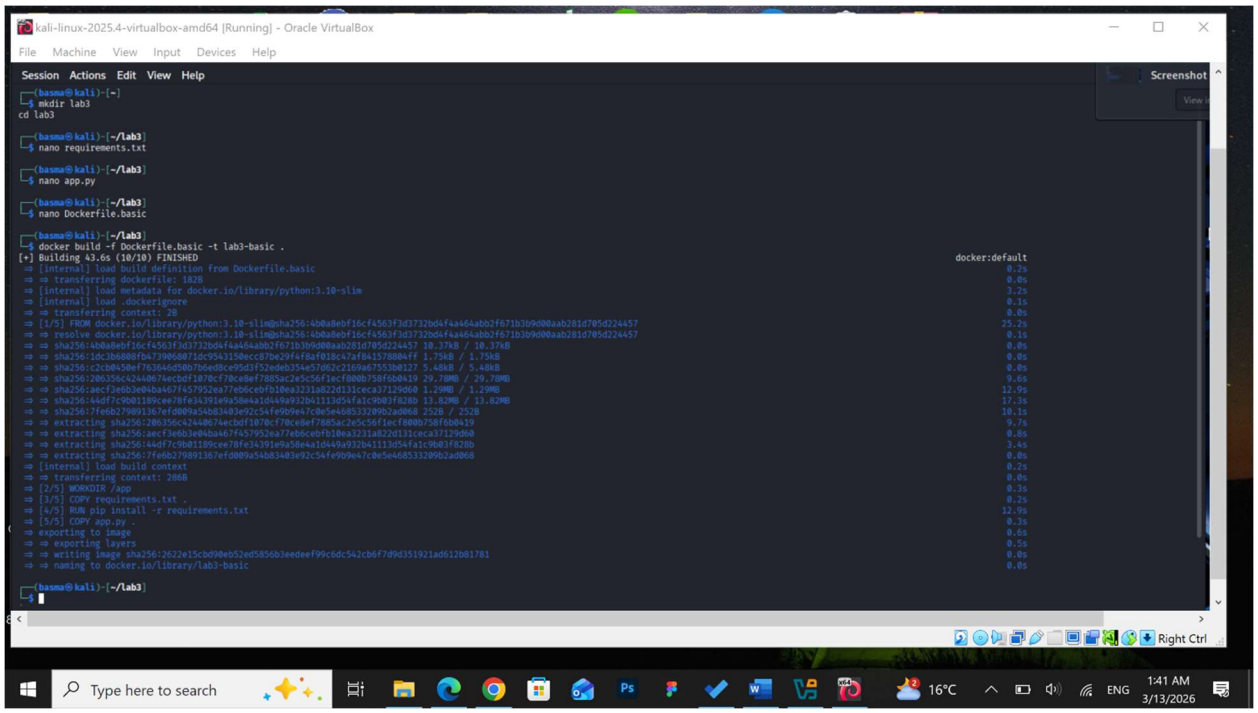
After:



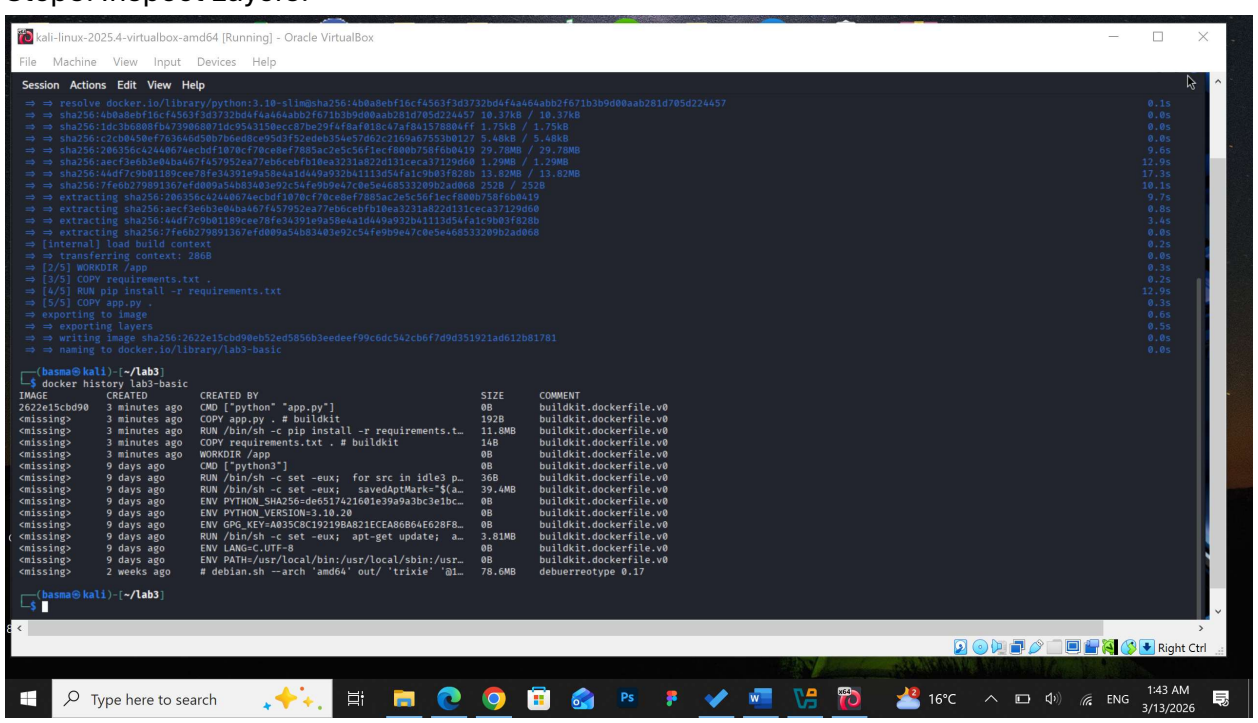
3. Steps of Building Basic Docker Image

Step1: Create Application Files:

Step2: Build the Image:

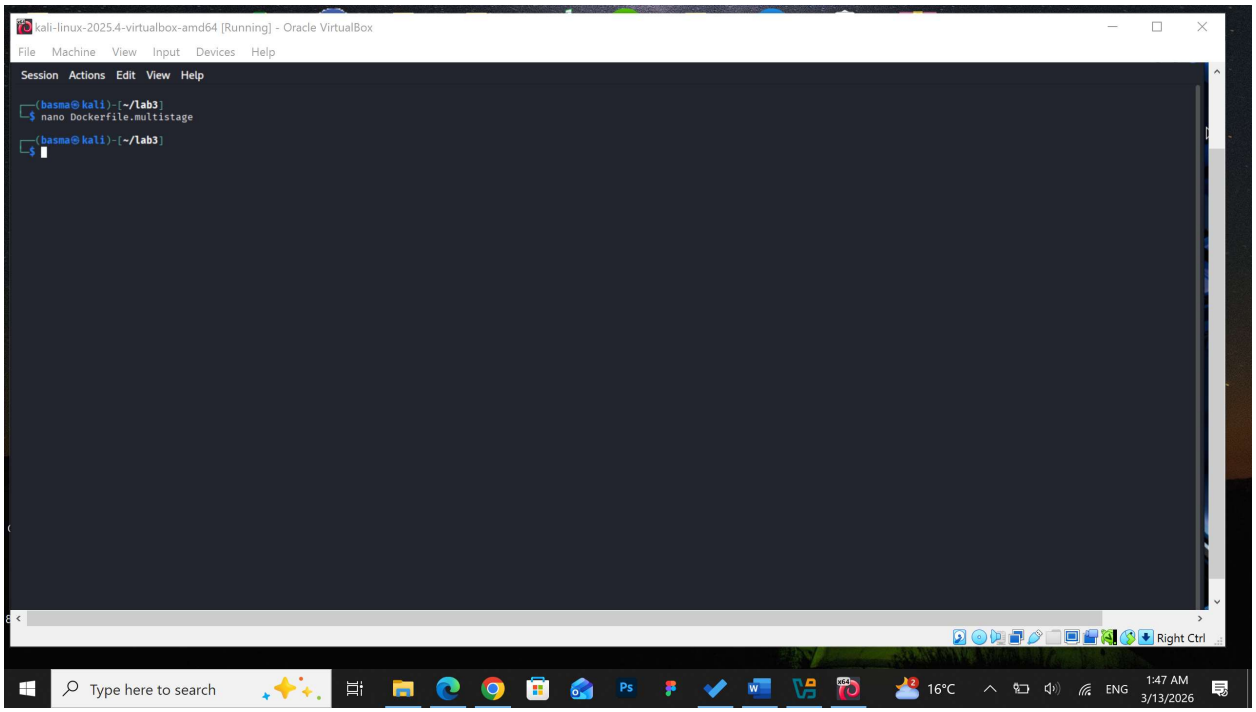


Step3: Inspect Layers:

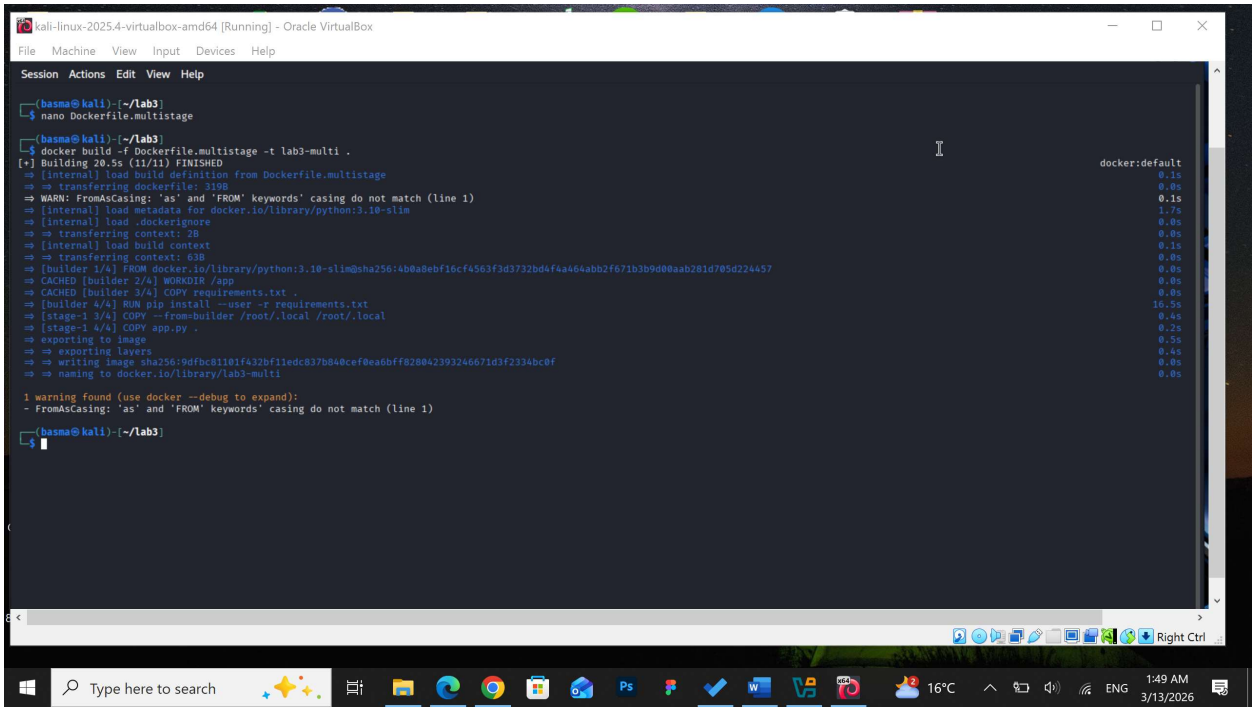


4. Steps of Multi-Stage Building of Docker Image

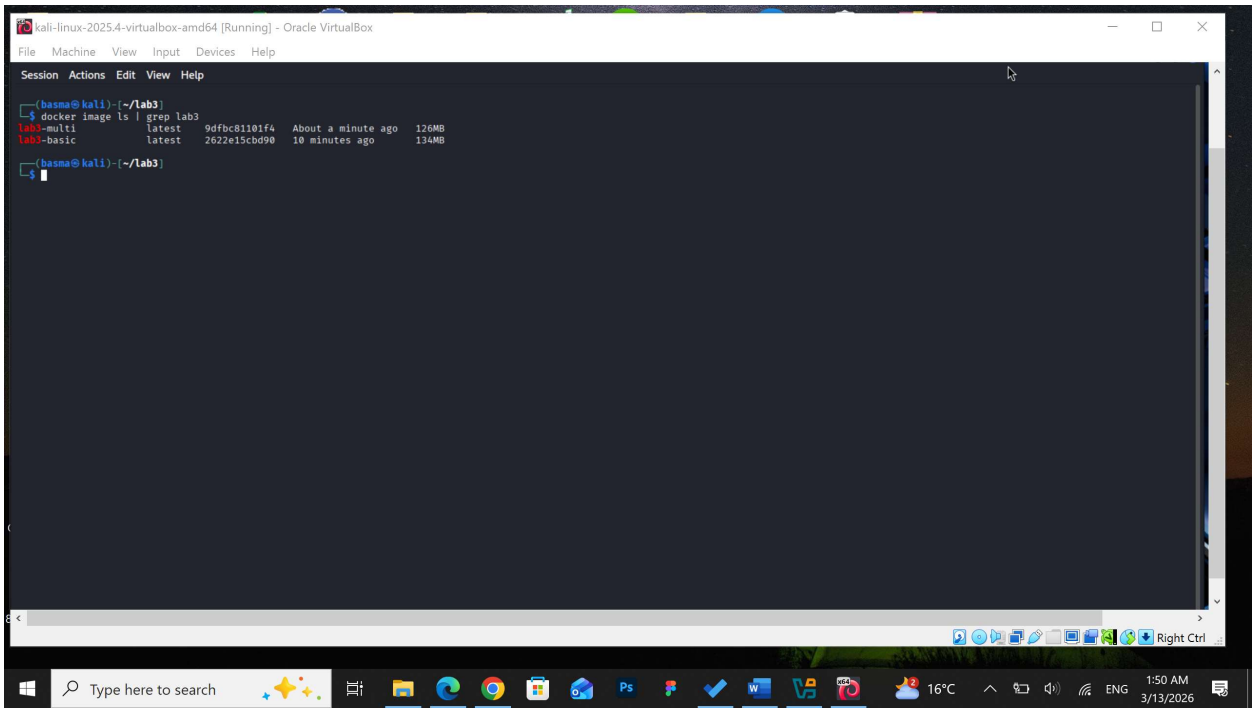
Step1: Create Multi-Stage Dockerfile:



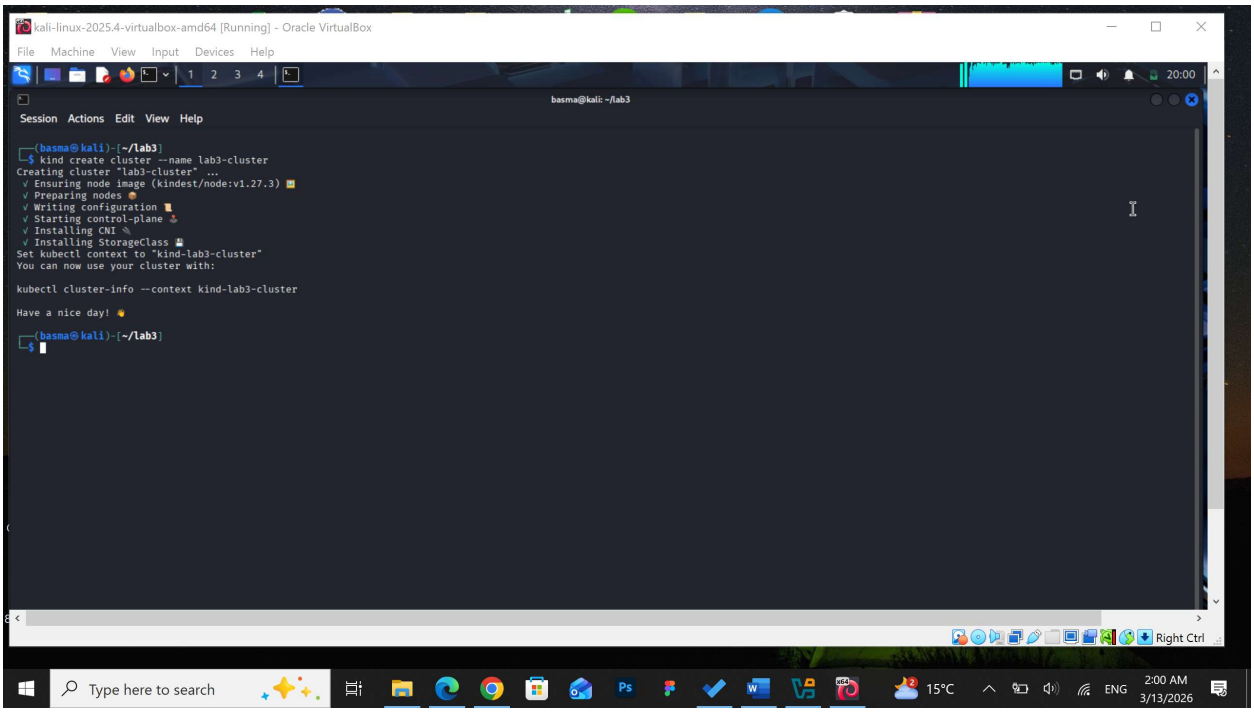
Step2: Build the Image:



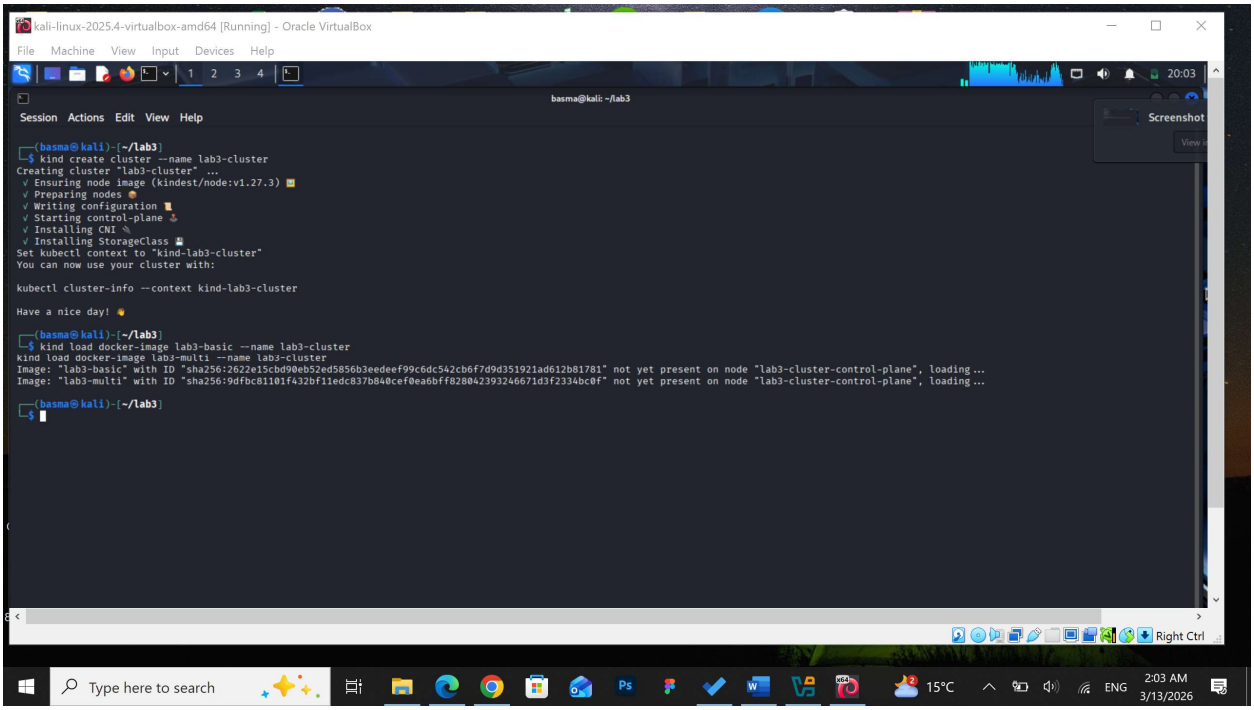
Step3: Compare Sizes:



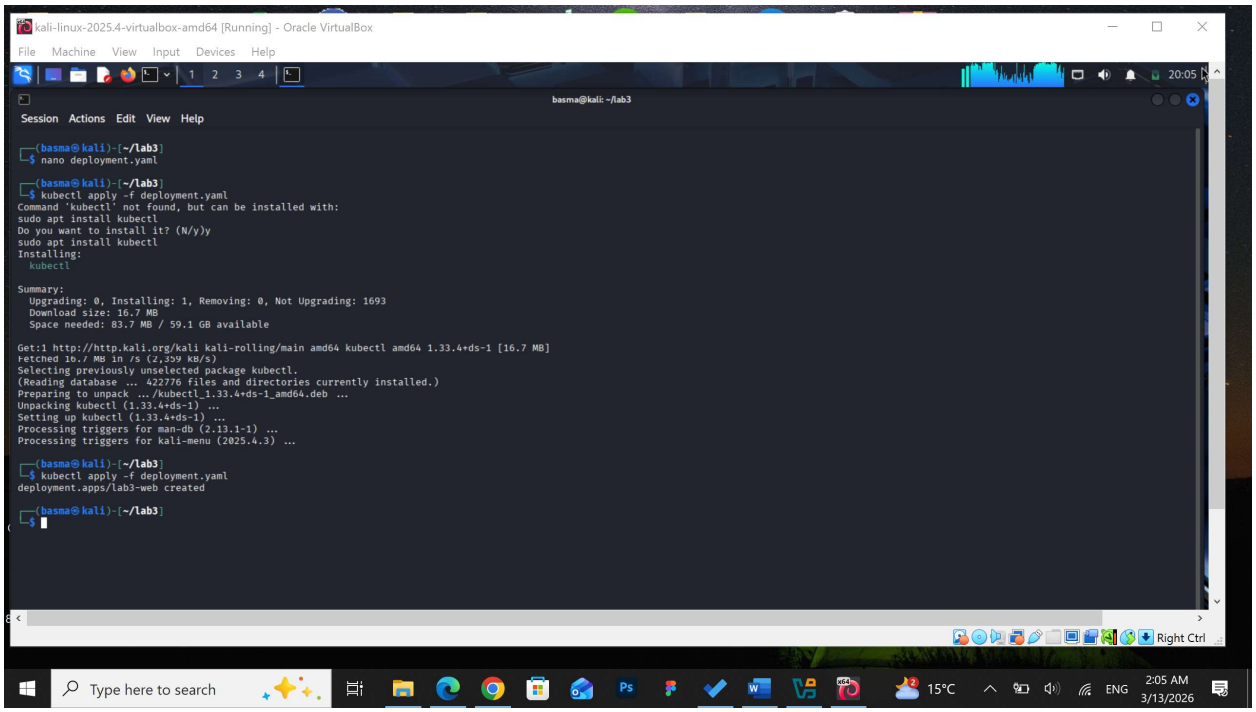
5. Create Local Cluster using ""kind""

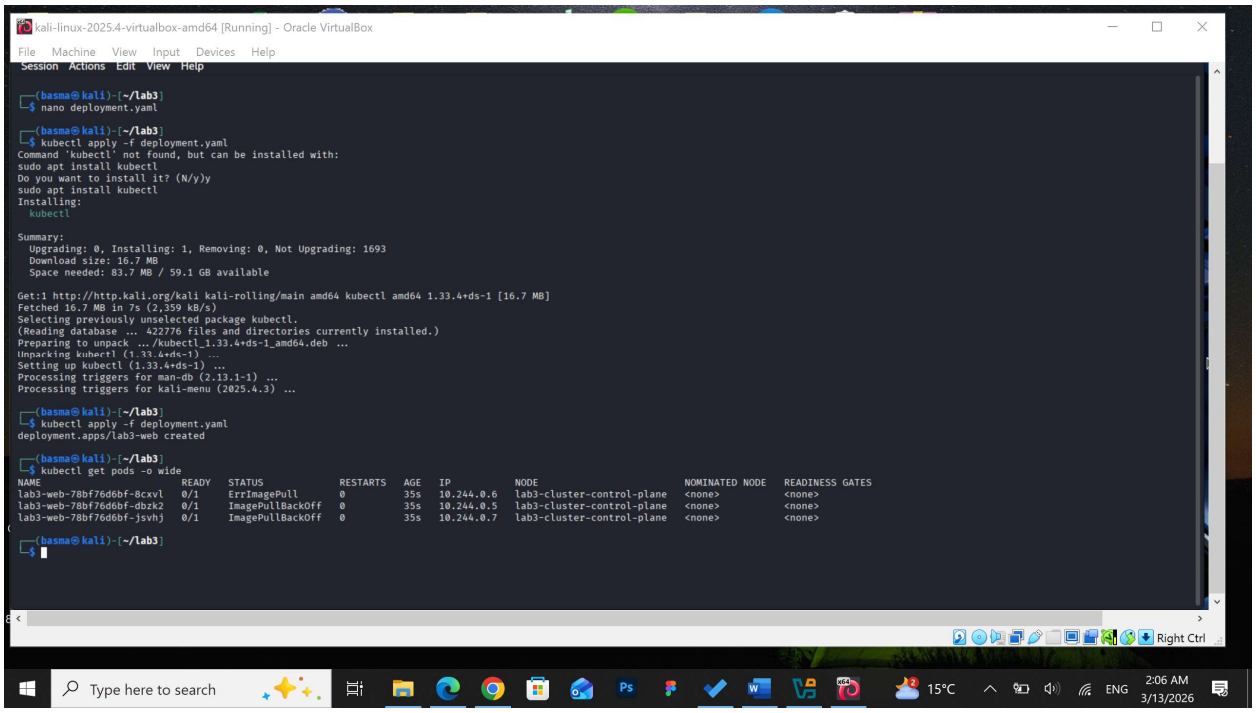


6. Copy images into the cluster



7. Get the deployment description





kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Session Actions Edit View Help

```
└─$ kubectl get pods -o wide
NAME                        READY   STATUS             RESTARTS   AGE   IP           NODE                      NOMINATED NODE   READINESS GATES
lab3-web-78bf76d6bf-8cxvl   0/1     ErrImagePull       0          35s   10.244.0.6   lab3-cluster-control-plane   <none>           <none>
lab3-web-78bf76d6bf-dbzk2   0/1     ImagePullBackOff   0          35s   10.244.0.5   lab3-cluster-control-plane   <none>           <none>
lab3-web-78bf76d6bf-jsvhj   0/1     ImagePullBackOff   0          35s   10.244.0.7   lab3-cluster-control-plane   <none>           <none>

┌──(basma㉿kali)-[~/lab3]
└─$ kubectl describe deployment lab3-web
Name:                   lab3-web
Namespace:              default
CreationTimestamp:      Thu, 12 Mar 2026 20:05:30 -0400
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=lab3-web
Replicas:               3 desired | 3 updated | 3 total | 0 available | 3 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=lab3-web
  Containers:
   web:
    Image:         lab3-multi
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      False   MinimumReplicasUnavailable
  Progressing    True    ReplicaSetUpdated
OldReplicaSets:  <none>
NewReplicaSet:   lab3-web-78bf76d6bf (3/3 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  89s   deployment-controller  Scaled up replica set lab3-web-78bf76d6bf to 3

┌──(basma㉿kali)-[~/lab3]
└─$
```

Right Ctrl

Type here to search

15°C ENG 2:07 AM 3/13/2026